

NORMAL-FORM COMPLEXITY OF A THREE-SHEETED KELLER COVER

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ABSTRACT. Let $K = X + Q + C$ be a degree-at-most-three Keller map in dimension n , with Q, C homogeneous of degrees two and three. If the coordinate span of C has dimension r , we give a collision-preserving cubic-homogeneous suspension in dimension $n + r + 1$. Applied to the public 11-variable degree-three descendant of the new Jacobian counterexample, where $r = 7$, this recovers the 19-variable construction independently announced by Harris Chan.

We determine the generic Jordan type of its nonlinear Jacobian as $(18, 1)$ and give a two-point certificate that its nilpotent flag moves. Symmetric doubling gives an explicit Hessian-nilpotent quartic in 38 variables, with generic Hessian type $(35, 2, 1)$. A two-parameter family of algebraic inverse rays shows

$$\Delta^m(Q^{m+1}) \neq 0 \quad \text{for every } m \geq 1,$$

and identifies the inverse singularities with intersections with the non-properness hypersurface. For the fixed 19-variable cubic tensor, we construct a 110-variable square-zero cubic-linear pairing and prove that every vector-Waring decomposition uses at least 52 cubes. Finally, an exact 13-term functional is a constant nonzero quartic obstruction across a natural 20-dimensional equivariant family of rank-six cubic compressions. The example-specific statements are computer-assisted and come with exact rational certificates.

Working computer-assisted proof draft. The recovered exact certificates have been rerun. Independent CAS reproduction and specialist review remain submission requirements.

1. INTRODUCTION

The classical reductions of Bass–Connell–Wright and Yagzhev make cubic-homogeneous maps central to the Jacobian conjecture [3]. Once a three-dimensional counterexample was announced [1], it therefore became natural to ask how economically that counterexample reaches the classical normal forms. We credit Akhil Mathew as the primary human source of the problem: Mathew prompted Levent Alpöge, Alpöge prompted Fable, Fable produced the example, and Alpöge announced it publicly [9].

Atkins-Turkish publicly supplied a degree-at-most-three map in 11 variables, with three explicit colliding points [2]. Its cubic coordinate space

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has dimension seven. The construction in section 2 consequently gives a cubic-homogeneous counterexample in

$$11 + 7 + 1 = 19$$

variables. Harris Chan independently announced this special sparse lift on July 23, 2026 [4]; we make no sole-priority claim for the 19-variable endpoint. The reusable contribution is the rank-sensitive $n + r + 1$ theorem.

The fixed 19-variable tensor has further structure. Its nonlinear Jacobian has generic Jordan type $(18, 1)$. Classical symmetric reduction [5] then produces a quartic potential in 38 variables. A 48-variable explicit Hessian witness was already public through the work of Santib   ez-Leal, starting from Thompson’s 24-variable cubic map [8, 7]. Our claims are the quantitative dimension 38, the exact Hessian Jordan type $(35, 2, 1)$, and the all-order Laplacian formula.

In the cubic-linear direction, we use the pairing formalism of Gorni and Zampieri [6]. The existence of such pairings and the square-zero mechanism are not new. For this particular tensor we prove the model-relative interval

$$52 \leq N_{\text{pair}} \leq 110.$$

Neither endpoint is asserted to be sharp for arbitrary cubic-homogeneous or cubic-linear counterexamples.

The main results are summarized as follows.

Theorem 1.1 (Rank-sensitive suspension). *Let $K = X + Q + C: \mathbb{A}^n \rightarrow \mathbb{A}^n$ be Keller over a characteristic-zero field, with Q quadratic homogeneous and C cubic homogeneous. Put*

$$r = \dim \text{span} \{C_1, \dots, C_n\}.$$

Then K admits a cubic-homogeneous Keller suspension in dimension $n+r+1$. Every collision of K lifts explicitly to a collision of the suspension.

Theorem 1.2 (The fixed 19-variable tensor). *For the tensor H of section 3:*

- (i) *$I + H$ is a noninjective cubic-homogeneous Keller map in 19 variables;*
- (ii) *the generic Jordan type of JH is $(18, 1)$;*
- (iii) *the associated 38-variable quartic $\mathcal{Q}$ is Hessian nilpotent, has generic Hessian type $(35, 2, 1)$, and violates Zhao’s eventual vanishing conclusion at every order; and*
- (iv) *the vector-Waring length of H is at least 52, while H has a full-rank square-zero pairing of length 110.*

Theorem 1.3 (Equivariant compression obstruction). *Let V be the space of quadratic source fields that preserve the weights; it has dimension 115. The locus in V killing five specified cubic output rows contains a natural affine slice of dimension 20. Each point on that slice has at most six linearly independent cubic coordinate functions. The same functional, supported on 13 terms, takes the value 1 on its quartic obstruction. Thus no point lifts*

through quartic order in the stated equivariant conjugacy model. The original 605-shift statement is a sparse corollary.

Theorem 1.3 is deliberately model-relative. It does not prove that 19 variables are globally minimal.

2. A RANK-SENSITIVE CUBIC SUSPENSION

Let k be a field of characteristic zero and write

$$K(X) = X + Q(X) + C(X),$$

where Q and C are homogeneous of degrees two and three. Choose a basis $q = (q_1, \dots, q_r)^T$ of the coordinate span of C , and a constant matrix $B \in M_{n \times r}(k)$ such that

$$C = Bq.$$

Introduce $w = (w_1, \dots, w_r)$ and define

$$\widetilde{K}(X, w) = (X + Q(X) + Bw, w - q(X)). \quad (2.1)$$

Lemma 2.1. *The map $\widetilde{K}$ is polynomially stably left-right equivalent to K .*

Proof. Set

$$S(X, w) = (X, w - q(X)), \quad T(U, V) = (U + BV, V).$$

Both are triangular polynomial automorphisms, and direct substitution gives

$$\widetilde{K} = T \circ (K \times \text{id}_{\mathbb{A}^r}) \circ S.$$

□

Adjoin a homogenizing coordinate t , and put

$$G(X, w, t) = (X + tQ(X) + t^2Bw, w - q(X), t). \quad (2.2)$$

Its nonlinear part is cubic homogeneous.

Proof of theorem 1.1. The Jacobian matrix in the X, w coordinates has block form

$$\begin{pmatrix} I + tJQ(X) & t^2B \\ -Jq(X) & I \end{pmatrix}.$$

The Schur complement gives

$$\begin{aligned} \det JG &= \det(I + tJQ(X) + t^2BJq(X)) \\ &= \det JK(tX) = 1. \end{aligned} \quad (2.3)$$

If $K(p) = K(p')$, then

$$\widehat{p} = (p, q(p), 1), \quad \widehat{p}' = (p', q(p'), 1)$$

are distinct and satisfy

$$G(\widehat{p}) = (K(p), 0, 1) = G(\widehat{p}').$$

Finally, write $G = I + H$. Since H is cubic homogeneous,

$$JH(sZ) = s^2JH(Z).$$

The identity $\det(I + JH) = 1$ therefore implies $\det(I + \lambda JH) = 1$ as a polynomial identity in λ . All nonconstant characteristic coefficients vanish, so JH is nilpotent. $\square$

Remark 2.2. For a fixed K , any factorization $C = B'q'$ through s scalar cubic coordinates has $s \geq r$. Thus r auxiliaries are optimal within the factorized suspension (2.1). This says nothing about other stable representatives of K or other homogenization models.

3. THE FIXED 19-VARIABLE TENSOR

We now work over $\mathbb{Q}$. In the coordinate order

$$X = (x, y, z, a, b, c, d, q, s, h, k),$$

let $\Phi = (\Phi_1, \dots, \Phi_{11})$ be the public map [2]:

$$\begin{aligned} \Phi_1 &= -ac - adz - 3ay^2 - 2az - cd^2 + d^2z - ds + 7dy^2 \\ &\quad + sxy + 3xyz + 4y^2 + z, \\ \Phi_2 &= -bc - bdz - 3by^2 - 2bz - 3cdx - dq + qxy \\ &\quad + 12xy^2 + 3xz + y, \\ \Phi_3 &= -hk - hxz + kx^2 - 3x^2y + 2x, \\ \Phi_4 &= a - d^2 + 2dxy, \quad \Phi_5 = b + 3x^2y, \\ \Phi_6 &= c + xyz + 3y^2 + 2z, \quad \Phi_7 = d - xy, \\ \Phi_8 &= bz + 3cx + q, \\ \Phi_9 &= s + az + cxy - xyz - 7y^2 + cd - dz, \\ \Phi_{10} &= h - x^2, \quad \Phi_{11} = k + xz. \end{aligned}$$

Its linear part $L = J\Phi(0)$ has determinant -2 . Normalize by

$$K = L^{-1}\Phi = X + Q + C.$$

The coordinate cubics

$$q_1 = C_1, \quad q_2 = C_2, \quad q_3 = C_3, \quad q_4 = C_4, \quad q_5 = C_5, \quad q_6 = C_6, \quad q_7 = C_9$$

are linearly independent, and the remaining coordinate cubics vanish. Thus $C = Bq$, where B includes these seven coordinates into $\mathbb{Q}^{11}$.

With $w = (w_1, \dots, w_7)$, define

$$H(X, w, t) = (tQ(X) + t^2Bw, -q(X), 0) \tag{3.1}$$

in the coordinate order

$$Z = (x, y, z, a, b, c, d, q, s, h, k, w_1, \dots, w_7, t).$$

Then $G = I + H$ is the suspension (2.2).

For reference, three colliding points of Φ are

$$\begin{aligned} p_1 &= (0, 0, -1/4, 0, 0, 1/2, 0, 0, 0, 0), \\ p_2 &= (1, -3/2, 13/2, -9/4, 9/2, -10, -3/2, 3/4, -153/8, 1, -13/2), \\ p_3 &= (-1, 3/2, 13/2, -9/4, -9/2, -10, -3/2, -3/4, -153/8, 1, 13/2). \end{aligned}$$

They all map to $(-1/4, 0, \dots, 0)$. Consequently

$$(p_j, q(p_j), 1), \quad j = 1, 2, 3,$$

are three colliding points of G .

3.1. Generic Jordan type. Let $A = JH$, and set

$$E = xy(c + 2z) + d^2z + 3dy^2.$$

Define two polynomial vectors

$$v_1 = 2tx e_h + 2tz e_k + (xk + zh)e_{w_1}, \quad (3.2)$$

$$v_2 = -txy e_a - t(dz + 3y^2)e_s - Ee_{w_3} + 2Ee_{w_6} + xyz e_{w_7}. \quad (3.3)$$

Proposition 3.1. *Over $\mathbb{Q}(Z)$, the matrix A has Jordan type $(18, 1)$.*

Proof. Exact differentiation gives

$$Av_1 = Av_2 = 0.$$

The vectors are generically independent, while the all-ones specialization of A has rank 17. Hence the generic kernel has dimension two.

Let e_t be the last coordinate vector. Exact iteration gives

$$A^{17}e_t = 18t^{15}x^9y^5z^2v_2 \neq 0. \quad (3.4)$$

By theorem 1.1, A is nilpotent. A nilpotent 19-by-19 matrix with a two-dimensional kernel has two Jordan blocks, and (3.4) forces the larger block to have size at least 18. Thus the partition is $(18, 1)$. $\square$

Remark 3.2. At the all-ones specialization the ranks of $A, A^2, \dots, A^{18}$ are

$$17, 16, \dots, 1, 0.$$

The vectors

$$e_t, Ae_t, \dots, A^{17}e_t, v_1$$

specialize to a basis with determinant 1952152956156672.

4. A RESTRICTED COMPRESSION OBSTRUCTION

Consider quadratic near-identity source conjugacies. At cubic order their action is

$$C \longmapsto C + [Q, P_2], \quad [Q, P_2] = JQ P_2 - JP_2 Q.$$

The shift

$$P_2 = -d^2e_a$$

lowers the cubic coordinate-span rank from seven to six. Its exact conjugate has quartic remainder

$$(D_4)_z = d^3z + 3d^2y^2, \quad (D_4)_a = x^2y^2, \quad (D_4)_c = -2d^3z - 6d^2y^2. \quad (4.1)$$

For a quartic vector field R , define

$$\begin{aligned} \Lambda_4(R) = & [x^2y^2]R_a + 4[x^2ac]R_a - \frac{20}{3}[x^2as]R_a + 9[x^2qk]R_a \\ & + 3[xchk]R_a - \frac{1}{2}[xd^2q]R_a - 7[xshk]R_a + \frac{4}{3}[y^2ah]R_a \\ & - \frac{1}{2}[ybch]R_a - [ydhk]R_a + \frac{8}{3}[za^2h]R_a + 25[aqhk]R_a - 2[dqhk]R_d. \end{aligned} \quad (4.2)$$

Proposition 4.1. *For every cubic vector field P_3 ,*

$$\Lambda_4([Q, P_3]) = 0, \quad \Lambda_4(D_4) = 1.$$

Thus $D_4 \notin \text{im}[Q, -]$.

Moreover, among the 605 shifts

$$P_2 = \lambda m e_j,$$

where m is a quadratic monomial not involving X_j , the shift $-d^2e_a$ is the unique one that lowers the cubic span to six.

Proof. The first statement is finite coefficient linear algebra. Evaluating (4.2) on all

$$11 \binom{13}{3} = 3146$$

cubic monomial vector fields gives zero, while direct evaluation on (4.1) gives one. The restricted homological matrix on the 13 displayed coefficient positions has rank 12 and augmented rank 13.

For the second statement, a fixed nonzero seven-by-seven minor of the cubic coefficient matrix is evaluated along each of the 605 directions. Of these, 583 have no exceptional parameter and 22 give 27 rational roots. Exact full-rank evaluation at every root leaves only $P_2 = -d^2e_a$. $\square$

Proposition 4.2. *Use the quadratic output corrections generated by the four outputs d, q, h, k , which have degree at most two after the conjugacy. Their ten quadratic products create a ten-dimensional quartic image. The z - and c -components of D_4 lie outside this image. The a -component has the unique preimage $-Y_d^2$, but applying it restores cubic coordinate-span rank seven.*

Proof. This is another exact coefficient-matrix calculation. The ten quartic images are independent. Solving against the three nonzero components in (4.1) gives the assertions. $\square$

5. THE 38-VARIABLE HESSIAN QUARTIC

Let $F(x) = x + H(x)$ be the 19-variable map. On $\mathbb{A}_x^{19} \times \mathbb{A}_y^{19}$, define the cotangent double

$$\Phi_F(x, y) = (F(x), JF(x)^T y).$$

Its Jacobian is block lower triangular, with diagonal blocks JF and JF^T , so $\det J\Phi_F = 1$.

Over $\mathbb{Q}(i)$, put

$$T(u, v) = \left(u + iv, \frac{u - iv}{2} \right).$$

The scalar block satisfies

$$TT^T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Therefore

$$\mathcal{G} = T^{-1}\Phi_F T = I + \nabla \mathcal{Q},$$

where

$$\mathcal{Q}(u, v) = \frac{1}{2}(u - iv)^T H(u + iv). \quad (5.1)$$

This is a homogeneous quartic with 724 nonzero monomials. If $F(p) = F(p')$, then

$$\Phi_F(p, 0) = \Phi_F(p', 0),$$

so $\mathcal{G}$ has an explicit transported collision.

Proposition 5.1. *The generic Jordan type of $\text{Hess } \mathcal{Q}$ is*

$$(35, 2, 1).$$

Proof. Before the linear twist, the nonlinear Jacobian is

$$N = \begin{pmatrix} A & 0 \\ \mathcal{L} & A^T \end{pmatrix}, \quad A = JH(x), \quad \mathcal{L} = \text{Hess}_x(y^T H(x)).$$

The Hessian of $\mathcal{Q}$ is similar to N .

The kernel vectors (3.2)–(3.3) satisfy

$$D^2 H[v_1, v_2] = D^2 H[v_2, v_2] = 0.$$

Since they span $\ker A$ generically, this implies $\mathcal{L}v_2 \in \text{im } A^T$. In addition to the two kernel vectors $(0, \ker A^T)$, the matrix N therefore has a kernel vector whose first component is v_2 . Hence $\text{rank } N \leq 35$. An exact all-ones specialization has rank 35.

For $k \geq 1$, the lower-left block of N^k is

$$\sum_{j=0}^{k-1} (A^T)^j \mathcal{L} A^{k-1-j}.$$

Since $A^{18} = 0$, the only possible term for $k = 35$ is $(A^T)^{17} \mathcal{L} A^{17}$. The image of A^{17} is spanned by v_2 , and $D^2 H[v_2, v_2] = 0$, so this term vanishes. Thus $N^{35} = 0$. At the all-ones specialization,

$$(N^{34})_{38,19} = 648 \neq 0.$$

The nilpotency index is 35 and the rank is 35, so there are three blocks. The remaining three dimensions split as $2 + 1$, proving the result. $\square$

5.1. An all-order Laplacian certificate. Let

$$\bar{Y} = (1/2, 1, 0, \dots, 0, 1) \in \mathbb{Q}^{19}, \quad \bar{W} = T^{-1}(\bar{Y}, 0) = \left(\frac{\bar{Y}}{2}, -\frac{i\bar{Y}}{2} \right).$$

An exact inverse branch of the original three-dimensional map, lifted through the stable reductions, gives

$$(F^{-1}(s\bar{Y}))_1 = \frac{1/\sqrt{1-s^4}-1}{s^3} = \sum_{r \geq 1} \frac{1}{4^r} \binom{2r}{r} s^{4r-3}. \quad (5.2)$$

Consequently

$$(\mathcal{G}^{-1}(s\bar{W}))_{u_1} = \frac{1}{2} \sum_{r \geq 1} \frac{1}{4^r} \binom{2r}{r} s^{4r-3}. \quad (5.3)$$

Put $P = -\mathcal{Q}$, so that $\mathcal{G} = I - \nabla P$. Zhao's inversion formula for a Hessian-nilpotent potential gives homogeneous inversion terms [10]

$$R_m = \frac{1}{2^m m! (m+1)!} \Delta^m(P^{m+1}).$$

The degree of ∇R_m is $2m + 3$. Comparing (5.3) at $m = 2r - 3$ gives:

Theorem 5.2. *For every $r \geq 2$,*

$$\partial_{u_1} \Delta^{2r-3}(\mathcal{Q}^{2r-2})(\bar{W}) = \frac{(2r-3)!(2r-2)!}{16} \binom{2r}{r} \neq 0.$$

In particular,

$$\partial_{u_1} \Delta(\mathcal{Q}^2)(\bar{W}) = \frac{3}{4}.$$

Proof. The exponent $m + 1 = 2r - 2$ is even, so $P^{m+1} = \mathcal{Q}^{m+1}$. Multiply the coefficient of s^{4r-3} in (5.3) by $2^m m! (m+1)!$. $\square$

Since $\mathcal{Q}$ is Hessian nilpotent, $\Delta^m(\mathcal{Q}^m) = 0$ for all $m \geq 1$, while theorem 5.2 gives infinitely many nonzero $\Delta^m(\mathcal{Q}^{m+1})$. This is an explicit failure of eventual vanishing in Zhao's framework.

6. A 52-TO-110 SQUARE-ZERO PAIRING INTERVAL

Let $V = \mathbb{Q}^{19}$, and regard

$$H \in V \otimes \text{Sym}^3(V^*).$$

Its *vector-Waring length* is the least R for which

$$H(z) = \sum_{\nu=1}^R b_\nu \ell_\nu(z)^3. \quad (6.1)$$

6.1. **The lower bound.** Let

$$W = \text{span} \{ \partial H_i / \partial z_j \} \subset \text{Sym}^2(V^*).$$

Every decomposition (6.1) gives

$$W \subset U := \text{span} \{ \ell_\nu^2 \}.$$

Exact row reduction gives

$$\dim W = 39. \tag{6.2}$$

In the coordinate order of section 3, the 12-dimensional subspace

$$\mathcal{A} = \langle z, c, q, s, k, w_1, \dots, w_7 \rangle$$

is a common zero block:

$$W|_{\mathcal{A}} = 0.$$

The coefficient system for a constant vector v satisfying $JH(z)v = 0$ has rank 19; a specified 19-by-19 minor has determinant 18. Thus the ℓ_ν span V^* , and their restrictions span $\mathcal{A}^*$. Twelve independent restrictions have twelve independent squares. It follows that

$$\dim(U/W) \geq 12, \quad \dim U \geq 51.$$

Proposition 6.1. *Every vector-Waring decomposition of H has length at least 52.*

Proof. Suppose a decomposition has length at most 51. Every preceding inequality is then an equality:

$$R = \dim U = 51, \quad W = \ker(U \rightarrow \text{Sym}^2(\mathcal{A}^*)).$$

Choose a basis $\alpha_1, \dots, \alpha_{12}$ of $\mathcal{A}^*$ among the restricted forms. If $\alpha = \sum c_j \alpha_j$ and α^2 lies in the span of the α_j^2 , then all $c_i c_j$ for $i \neq j$ vanish. Hence every restricted form is proportional to one α_j .

Choose a complement $V = \mathcal{B} \oplus \mathcal{A}$, with $\dim \mathcal{B} = 7$, and let

$$\mathcal{C} = \text{pr}_{\mathcal{B}^* \otimes \mathcal{A}^*}(W).$$

The equality description forces $\mathcal{C}$ to be invariant under the twelve coordinate projectors in the α_j -basis. Therefore its right stabilizer algebra

$$\mathcal{E} = \{ T \in \text{End}(\mathcal{A}^*) : (I \otimes T)\mathcal{C} \subset \mathcal{C} \}$$

contains a conjugate of the full diagonal algebra. Its commutant can then contain no nonzero nilpotent.

Exact rational computation gives

$$\dim \mathcal{C} = 24, \quad \dim \mathcal{E} = 62, \quad \dim \mathcal{E}' = 7,$$

and $\mathcal{E}'$ contains the matrix unit $E_{5,0}$, with indices numbered $0, \dots, 11$. This element is nonzero and square-zero, a contradiction. $\square$

6.2. The 110-variable upper construction. The supplement contains rational matrices

$$B \in M_{19 \times 110}(\mathbb{Q}), \quad D \in M_{110 \times 19}(\mathbb{Q})$$

satisfying

$$\text{rank } B = \text{rank } D = 19, \quad BD = 0, \quad H(z) = B(Dz)^{*3}. \quad (6.3)$$

Put $A = DB$ and

$$F_A(W) = W + (AW)^{*3}.$$

Then $A^2 = 0$ and $\text{rank } A = 19$.

Proposition 6.2. *The map $F_A: \mathbb{A}^{110} \rightarrow \mathbb{A}^{110}$ is a noninjective Keller map. It is Gorni–Zampieri paired with $G = I + H$.*

Proof. Choose a right inverse C of B . The exact data give $AC = D$, and

$$BF_A(Cz) = z + B(Dz)^{*3} = G(z).$$

By Sylvester’s determinant identity,

$$\begin{aligned} \det JF_A(W) &= \det \left(I_{110} + 3 \text{diag}((AW)^2)DB \right) \\ &= \det \left(I_{19} + 3B \text{diag}((DBW)^2)D \right) \\ &= \det JG(BW) = 1. \end{aligned}$$

If $G(z_1) = G(z_2)$, put $u = Cz_1$, $v = Cz_2$, and $\delta = F_A(u) - F_A(v)$. Then $B\delta = 0$, hence $A\delta = DB\delta = 0$. Thus

$$F_A(v + \delta) = F_A(v) + \delta = F_A(u).$$

The exact collision certificate verifies $u \neq v + \delta$. $\square$

Corollary 6.3. *For the fixed tensor H , the minimum length of a full-rank square-zero pairing satisfies*

$$52 \leq N_{\text{pair}} \leq 110.$$

Proof. The lower bound is proposition 6.1; every such pairing is a vector-Waring decomposition. The upper bound is proposition 6.2. $\square$

7. EXACT COMPUTATION AND REPRODUCIBILITY

All example-specific calculations use exact rational arithmetic. The fresh audit reran:

- (1) the public-input-to-19D collision and nilpotency certificate;
- (2) the rank-seven factorization and Schur-complement identities;
- (3) both kernel vectors, the length-18 Jordan chain, and the degeneration certificate;
- (4) the 38D gradient identity, Hessian rank and index, inverse branch, and first Laplacian value;
- (5) the 110D matrices, square-zero identity, pairing identity, Keller determinant transfer, and collision;

- (6) the 52-cube lower-bound matrices and equality obstruction;
- (7) all 3,146 monomial tests for Λ_4 ;
- (8) all 605 triangular monomial shifts and the ten target-cleanup directions; and
- (9) a separate verifier using only Python rational fractions for the small rank, minor, circuit, and commutant certificates.

The regenerated 724-term quartic is byte-for-byte identical to the recovered output. The 110D pairing JSON and the serialized small certificates also match their archived inputs exactly.

The computations do not establish novelty, and the several SymPy scripts are not independent CAS implementations. The Fraction-only checker is independent of SymPy but consumes serialized certificate matrices. A public release should additionally reproduce the main identities in Singular, Macaulay2, Sage, or Magma.

8. QUESTIONS

Question 8.1 (Minimum homogeneous dimension). *What is the minimum dimension of a cubic-homogeneous counterexample? The fixed 19-variable construction gives an upper bound, while the known low-dimensional classifications and the five-dimensional restrictions in appendix A leave a substantial gap.*

Question 8.2. *Can one improve the rank-sensitive complexity*

$$\dim K + \dim \text{span} \{(K_3)_i\}$$

by choosing a different degree-at-most-three stable representative?

Question 8.3 (Intrinsic tensor and pairing ranks). *What are the exact vector-Waring length, pairing rank, and square-zero realization rank of the fixed 19-variable cubic tensor? In particular, can the present vector-Waring interval between 52 and 110 be closed without committing to one reduction algorithm?*

Question 8.4. *Is there a conceptual explanation for the Jordan types $(18, 1)$ and $(35, 2, 1)$, independent of coefficient calculation?*

APPENDIX A. THE FIVE-DIMENSIONAL FRONTIER

The body studies explicit descendants in dimensions 19, 38, and 110. The opposite optimization problem begins in dimension five: cubic-homogeneous Keller maps in dimensions three and four are known to be invertible, so five is the first dimension in which a cubic-homogeneous counterexample could occur. This appendix records exact restrictions and a surviving line-level family. Nothing here constructs a five-dimensional counterexample.

A.1. Collision pencils. Let

$$F(x) = x + H(x), \quad H(x) = T(x, x, x),$$

where $T: V^3 \rightarrow V$ is symmetric and $\dim V = 5$. Suppose

$$F(v + u) = F(v)$$

with u, v linearly independent. On $\mathbb{P}(\langle u, v \rangle)$, write

$$M(s, t) = JH(su + tv) = s^2A + stB + t^2C,$$

where

$$A = JH(u), \quad B = 6T(u, v, -), \quad C = JH(v).$$

Symmetry of T and the collision give

$$Bu = 2Av, \quad Bv = 2Cu, \quad u + \frac{1}{3}Au + Av + Cu = 0. \quad (\text{A.1})$$

Conversely, the first two identities are sufficient to extend the plane data to a symmetric trilinear tensor. They do not ensure that the global Jacobian is nilpotent away from the plane.

Proposition A.1 (Three generic Jordan strata). *For a five-dimensional cubic-homogeneous counterexample, the generic nilpotent Jordan type of JH must be one of*

$$(5), \quad (4, 1), \quad (3, 2).$$

Proof. Known low-rank results exclude generic rank at most two in a five-dimensional counterexample, while nilpotence gives rank at most four. The nilpotent partitions of five having rank three or four are precisely the three displayed partitions. $\square$

A.2. A corrected sparse exclusion. Consider the square-zero two-matrix family

$$H_{C,Q}(z) = C \left((Qz)^{*3} - Qz^{*3} \right).$$

After a signed-permutation normalization, let $Q = I + R$, where R has exactly three active rows and exactly one nonzero off-diagonal entry in each active row. Impose the coordinate collision

$$(I + H_{C,Q})(e_0) = (I + H_{C,Q})(e_1).$$

Theorem A.2 (Three-row sparse exclusion). *No member of this coordinate-collision ansatz whose three directed support edges collectively use all five vertices is Keller.*

Proof. For one edge $i \rightarrow j$ with coefficient r , its cubic contribution is

$$U = (z_i + rz_j)^3 - z_i^3 - rz_j^3.$$

If b is the corresponding column of the compressed coefficient matrix, the z_i^2 and $z_i z_j$ coefficients in its contribution to the trace are

$$3rb_j, \quad 6rb_i + 6r^2b_j.$$

Thus the trace equation forces $b_i = b_j = 0$.

There are 180 three-edge patterns with distinct source rows. The collision and trace constraints leave 30, forming three orbits under the evident vertex symmetries:

$$\begin{array}{l|l} A & (0 \rightarrow 2, 3 \rightarrow 0, 4 \rightarrow 1), \\ B & (2 \rightarrow 0, 3 \rightarrow 0, 4 \rightarrow 1), \\ C & (2 \rightarrow 0, 3 \rightarrow 1, 4 \rightarrow 2). \end{array}$$

Writing the edge coefficients as α, β, γ , exact elimination of the collision and trace equations leaves, respectively, the following nonzero coefficients in $\text{tr}(JH/3)^2$:

$$\frac{18}{\beta^2 - 1} \quad \text{in orbit } A, \quad 18 \quad \text{in orbit } C.$$

In the generic branch of orbit B , two coefficients force both

$$B_{1,1} = 0, \quad (\beta^3 - \beta)B_{1,1} = 1.$$

The branches $\alpha = \pm 1$ and $\beta = \pm 1$ instead leave

$$\frac{36\beta}{\beta^2 - 1} \quad \text{or} \quad \frac{36\alpha}{\alpha^2 - 1}.$$

Every branch is contradictory. $\square$

Remark A.3 (Scope correction). The theorem is not the blanket statement that every support- ≤ 4 five-dimensional ansatz fails. Its coordinate-collision, three-row, and one-edge-per-row hypotheses are essential. An earlier finite-field experiment using only coefficients $0, \pm 1$ was vacuous for this collision and is not evidence for the theorem; the exact characteristic-zero argument above is the proof.

A.3. Everywhere-regular Jordan type (5). Assume $M(s, t)$ has Jordan type (5) at every point of $\mathbb{P}^1$. Regard it as

$$M: \mathcal{O}_{\mathbb{P}^1}^5 \longrightarrow \mathcal{O}_{\mathbb{P}^1}^5(2).$$

For $K_i = \ker M^i$ and $L_i = K_i/K_{i-1}$, the induced maps

$$L_i \longrightarrow L_{i-1}(2)$$

are isomorphisms. Since the degrees sum to zero,

$$(\deg L_1, \dots, \deg L_5) = (-4, -2, 0, 2, 4). \quad (\text{A.2})$$

In particular, the right kernel is an $\mathcal{O}(-4)$ -subbundle of $\mathcal{O}^5$.

If r is the coordinate-span dimension of a quartic kernel generator, the quadratic-syzygy calculation leaves only

r	possible splitting
3	(2, 2)
4	(1, 1, 2)
5	(1, 1, 1, 1),

while $r = 2$ is impossible.

In the full-span chart, normalize the kernel to the rational normal quartic

$$k(s, t) = (s^4, s^3t, s^2t^2, st^3, t^4)^T.$$

Its linear syzygy matrix is

$$K(s, t) = \begin{pmatrix} -t & s & 0 & 0 & 0 \\ 0 & -t & s & 0 & 0 \\ 0 & 0 & -t & s & 0 \\ 0 & 0 & 0 & -t & s \end{pmatrix}.$$

The linear lifting system for the regular nilpotent quotient has rank 48 in 56 unknowns and hence an eight-dimensional solution space. On the open set where the resulting constant 4×4 matrix T is invertible, the pencil factors as

$$M(s, t) = R(s, t)T^{-1}K(s, t)$$

and satisfies $M^5 = 0$.

Theorem A.4 (A smooth characteristic-zero collision-line family). *The everywhere-regular type-(5) equations (A.1) have a smooth one-dimensional family in characteristic zero.*

Exact computer-assisted existence proof. In the eight-parameter full-kernel chart, the following point over $\mathbb{F}_{11}$ satisfies the integrability and collision equations:

$$a = (8, 7, 1, 7, 2, 9, 0, 1), \\ u = (7, 6, 3, 10, 4), \quad v = (8, 9, 7, 9, 1).$$

At this point $\det T = 1$. The gcd of the 4×4 minors of $M(s, t)$ is one in $\mathbb{F}_{11}[s, t]$, so the pencil is geometrically regular on $\mathbb{P}_{\mathbb{F}_{11}}^1$.

After fixing $a_7 = 1$ and one coordinate of v , the ten integrability and five collision equations are functions of 16 variables. Their Jacobian has rank 15; a specified 15×15 minor has determinant 3 modulo 11. The multivariate Hensel lemma therefore gives a one-dimensional smooth characteristic-zero lift. The archived exact script checks the lift through modulus

$$11^{12} = 3138428376721.$$

Geometric regularity is open, so it persists on the lifted family. $\square$

A.4. A global rank-drop obstruction. Let $M(x)$ be a homogeneous quadratic nilpotent 5×5 matrix on $\mathbb{P}^4$, generically of type (5), and put

$$D = \{[x] \in \mathbb{P}^4 : \text{rank } M(x) \leq 3\}.$$

Theorem A.5 (Codimension-two rank drop). *One has*

$$\text{codim}_{\mathbb{P}^4} D \leq 2.$$

If M extends an everywhere-regular collision line, then

$$\text{codim}_{\mathbb{P}^4} D = 2$$

and D is disjoint from that line.

Proof. Suppose $\text{codim } D \geq 3$ and put $U = \mathbb{P}^4 \setminus D$. The regular kernel filtration on U has line quotients

$$L_i \simeq \mathcal{O}_U(-4 + 2(i - 1))$$

by the same calculation as (A.2). Removing a codimension-at-least-three set preserves Pic and CH^2 . Multiplicativity of Chern classes in the filtration of $\mathcal{O}_U^5$ would give

$$0 = c_2(\mathcal{O}_U^5) = \sum_{0 \leq i < j \leq 4} (-4 + 2i)(-4 + 2j)h^2 = -20h^2,$$

contradicting $h^2 \neq 0$ in $\text{CH}^2(U)$.

An everywhere-regular collision line is disjoint from D . A divisor in $\mathbb{P}^4$ cannot be disjoint from a projective line, so D cannot have codimension one. The preceding bound then forces codimension two. $\square$

Remark A.6 (The remaining problem). Theorem A.4 supplies only necessary plane data. A global extension still has 110 free symmetric tensor entries and must satisfy all coefficients of

$$\text{tr}(JH(x)^j) = 0, \quad j = 1, \dots, 5.$$

Every successful extension must also acquire the codimension-two rank-drop surface of theorem A.5. Constructing or excluding such an extension remains open.

APPENDIX B. ONE COVER IN SEVERAL NORMAL FORMS

The reductions below do more than transport a collision: after adjoining purely transcendental variables, they preserve the function-field extension induced by the original map. Thus the cubic, symmetric, and square-zero descendants are presentations of the same generically three-sheeted cover.

Proposition B.1 (Rees presentation of the rank-sensitive suspension). *Let $K = X + Q + C$, let $C = Bq$, and let*

$$G(X, w, t) = (X + tQ(X) + t^2Bw, w - q(X), t).$$

Define triangular automorphisms

$$S(X, v, t) = (X, v + q(X), t), \quad T(Y, v, t) = (Y - t^2Bv, v, t).$$

Then

$$T \circ G \circ S = (K_t \times \text{id})(X, v, t), \quad K_t(X) = X + tQ(X) + t^2C(X) = t^{-1}K(tX).$$

Consequently, after inverting t , the function-field extension induced by G is obtained from that induced by K by adjoining independent transcendental variables. In particular, generic degree and geometric monodromy are preserved.

Proof. Direct substitution gives

$$G(X, v + q(X), t) = (X + tQ(X) + t^2Bv + t^2C(X), v, t),$$

and the target shear T removes t^2Bv . Over $k(t)$ the change of variables $U = tX$ identifies K_t with K . $\square$

Proposition B.2 (The symmetric double is a stable presentation). *Let $F = X + H$ be Keller and put*

$$\Phi_F(x, y) = (F(x), JF(x)^T y).$$

Then the polynomial automorphism

$$S_F(x, y) = (x, JF(x)^T y)$$

satisfies

$$\Phi_F = (F \times \text{id}) \circ S_F.$$

Hence the symmetric double, and therefore its linear gradient twist, induces the same function-field extension as F after adjoining $\dim F$ independent variables.

Proof. Since $\det JF = 1$, the inverse matrix JF^{-T} has polynomial entries, so S_F is a polynomial automorphism. $\square$

Proposition B.3 (A full-rank square-zero pairing is stably right-equivalent). *Suppose*

$$H(z) = B(Dz)^{*3}, \quad BD = 0,$$

*where $B : W \rightarrow V$ is surjective and $D : V \rightarrow W$ is injective. Put $A = DB$ and $F_A(w) = w + (Aw)^{*3}$. Choose a right inverse C of B and write $W = C(V) \oplus E$, where $E = \ker B$. Set*

$$\rho(z) = (Dz)^{*3} - CH(z) \in E.$$

In the coordinates $w = Cz + \eta$, one has

$$F_A(z, \eta) = (G(z), \eta + \rho(z)), \quad G = I + H.$$

Precomposition by $(z, \eta) \mapsto (z, \eta - \rho(z))$ therefore changes F_A into $G \times \text{id}_E$.

B.1. The quotient cubic and its monodromy. For the original three-variable map, write its target coordinates as (ξ, η, ζ) . The source has the grading $\text{wt}(x, y, z) = (1, -1, -2)$, and the target weights are $(-2, -1, 1)$. Put

$$\nu = xy, \quad v = x^2z, \quad r = 1 + \nu, \quad c = 2 - 3\nu - v, \quad q = cr,$$

and form the target invariants

$$A = \xi\zeta^2, \quad B = \eta\zeta.$$

A direct calculation gives

$$A = q^2 + cq - q^3, \quad B = 4q + 2c - 3q^2.$$

Eliminating c yields the cubic

$$q^3 - 2q^2 + Bq - 2A = 0. \tag{B.1}$$

Its discriminant is

$$-4(27A^2 - 18AB + 16A + B^3 - B^2) = -4\zeta^2 \mathcal{N}(\xi, \eta, \zeta),$$

where

$$\mathcal{N}(\xi, \eta, \zeta) = 27\xi^2\zeta^2 - 18\xi\eta\zeta + 16\xi + \eta^3\zeta - \eta^2. \quad (\text{B.2})$$

The quadratic polynomial in A inside the discriminant has discriminant $-4(3B - 4)^3$, hence is not a square over $\mathbb{Q}(B)$. Together with the generic irreducibility of (B.1), this gives geometric monodromy S_3 .

APPENDIX C. A TWO-PARAMETER FAMILY OF ALL-ORDER INVERSE RAYS

Let e_x, e_y, e_{w_1}, e_t refer to the indicated coordinates of the 19-variable cubic map, and for $\alpha, \beta \in \mathbb{Q}$ put

$$Y_{\alpha, \beta} = \alpha e_x + e_y + \beta e_{w_1} + e_t.$$

Let $\hat{G}_0^{-1}$ denote the formal inverse branch at the origin.

Theorem C.1 (Exact inverse-ray family). *For all α, β ,*

$$(\hat{G}_0^{-1}(sY_{\alpha, \beta}))_x = \frac{(1 - 2\alpha s^4 + 2\beta s^6)^{-1/2} - 1}{s^3},$$

where the square root is the branch with constant term 1.

Proof. The t -coordinate gives $t = s$, and the w -coordinates give $w = q(X) + s\beta e_1$. Therefore

$$K(sX) = s^2(\alpha e_x + e_y) - s^4\beta e_x.$$

After applying the linear normalization and eliminating the stable auxiliary coordinates, this is the original three-variable target curve

$$(\xi, \eta, \zeta) = (0, s^2, 2\alpha s^2 - 2\beta s^4).$$

On the local branch above $\xi = 0$, write

$$R = (1 + xy)^2 z + y^2(4 + 3xy).$$

The equation $\xi = (1 + xy)R = 0$ gives $R = 0$, and then $\eta = y = s^2$. With $r = 1 + xy$, the third component simplifies to

$$\zeta = \frac{1 - r^{-2}}{y}.$$

Thus

$$r = (1 - y\zeta)^{-1/2} = (1 - 2\alpha s^4 + 2\beta s^6)^{-1/2},$$

and the original x -coordinate is $(r - 1)/s^2$. Since the original source is sX , division by s gives the formula. $\square$

The same calculation explains the singularities: along this curve,

$$\mathcal{N}(0, s^2, 2\alpha s^2 - 2\beta s^4) = -s^4(1 - 2\alpha s^4 + 2\beta s^6).$$

Thus the algebraic singularities of the inverse ray are precisely its intersections with the discriminant/nonproperness hypersurface.

Let $\mathcal{G} = I + \nabla \mathcal{Q}$ be the 38-variable gradient descendant, and put

$$W_{\alpha,\beta} = T^{-1}(Y_{\alpha,\beta}, 0).$$

For $n \geq 0$ define

$$c_n(\alpha, \beta) = [u^n](1 - 2\alpha u^2 - 2\beta u^3)^{-1/2}.$$

For $n \geq 2$ one has the finite sum

$$c_n(\alpha, \beta) = \sum_{k=\lceil n/3 \rceil}^{\lfloor n/2 \rfloor} 2^{-k} \binom{2k}{k} \binom{k}{n-2k} \alpha^{3k-n} \beta^{n-2k}. \quad (\text{C.1})$$

In particular, $c_n(\alpha, \beta) > 0$ for positive rational α, β .

Theorem C.2 (Nonvanishing at every Laplacian order). *For every $m \geq 1$ and every positive rational α, β ,*

$$\partial_{u_1} \Delta^m(\mathcal{Q}^{m+1})(W_{\alpha,\beta}) = 2^{m-1} m!(m+1)! c_{m+3}(\alpha, \beta) > 0.$$

Thus the same quartic violates the eventual-vanishing conclusion at every order, not merely along an infinite parity class.

Proof. The first coordinate of the inverse gradient ray is half the first coordinate in Theorem C.1. If

$$(1 - 2\alpha u^2 + 2\beta u^3)^{-1/2} = \sum d_n u^n,$$

then $d_n = (-1)^n c_n(\alpha, \beta)$. Compare the coefficient of s^{2m+3} with Zhao's inversion formula for $P = -\mathcal{Q}$:

$$R_m = \frac{1}{2^m m!(m+1)!} \Delta^m(P^{m+1}).$$

The two signs cancel, giving the displayed positive formula. $\square$

The central-binomial formula in the earlier draft is the boundary case $(\alpha, \beta) = (1/2, 0)$. Turning on β fills the parity gaps.

APPENDIX D. THE NILPOTENT FLAG MOVES

Let $A(Z) = JH(Z)$ for the 19-variable cubic tensor.

Proposition D.1 (A two-point obstruction to strong nilpotence). *Although $A(Z)$ is nilpotent for every Z , the family is not strongly nilpotent and is not simultaneously strictly triangularizable. Indeed,*

$$A(e_d)A(e_t)|_{\langle e_{w_3}, e_{w_6} \rangle} = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix},$$

and all other entries of the product vanish. Hence

$$\chi_{A(e_d)A(e_t)}(\lambda) = \lambda^{18}(\lambda + 3).$$

This gives a concrete form of the distinction between pointwise nilpotence and a common nilpotent flag: the generic matrix is almost regular nilpotent, but its Jordan flag twists with the base point.

D.1. A self-dual extension lemma.

Lemma D.2 (Isotropic terminal vector). *Let A have Jordan type $(m, 1)$, let $\ker A = \langle v_1, v_2 \rangle$, and suppose $\operatorname{im} A^{m-1} = \langle v_2 \rangle$. Let $L = L^T$ and*

$$N = \begin{pmatrix} A & 0 \\ L & A^T \end{pmatrix}.$$

Define the symmetric form

$$\beta(u, v) = u^T L v \quad (u, v \in \ker A).$$

If β has rank one with radical $\langle v_2 \rangle$, if $N^{2m-2} \neq 0$, and if $\beta(v_2, v_2) = 0$, then N has Jordan type $(2m-1, 2, 1)$.

Proof. Projection of $\ker N$ to its first factor identifies its image with $\operatorname{rad} \beta$, and each fiber is $\ker A^T$. Thus $\dim \ker N = 2 + 1 = 3$. The lower-left block of N^{2m-1} is the sole possible term

$$(A^T)^{m-1} L A^{m-1},$$

which vanishes because the terminal line is isotropic. The preceding power does not vanish:

$$N^{2m-2} \neq 0.$$

Thus the nilpotency index is $2m-1$. The operator acts on a space of dimension $2m+2$ and has three blocks. Its largest block has size $2m-1$, so its partition is $(2m-1, 2, 1)$. $\square$

For the present tensor, exact differentiation gives

$$D^2 H[v_1, v_1] = -4t^3 x z e_x, \quad D^2 H[v_1, v_2] = D^2 H[v_2, v_2] = 0.$$

Hence the form induced by $L = \operatorname{Hess}_x(y^T H(x))$ has rank one on $\ker A$, with radical $\langle v_2 \rangle$. The previously verified nonzero entry of N^{34} then applies Lemma D.2 with $m = 18$ and explains the partition $(35, 2, 1)$ as a one-box degeneration of maximal chain gluing.

APPENDIX E. A ROBUST EQUIVARIANT COMPRESSION OBSTRUCTION

The normalized 11-variable map is equivariant for the weights

$$(1, -1, -2, 0, 1, -2, 0, -1, -2, 2, -1)$$

on $(x, y, z, a, b, c, d, q, s, h, k)$. Let $\mathfrak{X}_2^{(0)}$ be the space of weight-preserving quadratic vector fields; exact counting gives $\dim \mathfrak{X}_2^{(0)} = 115$.

For a quadratic formal coordinate change $I + P$, the cubic jet is $C + [Q, P]$. Its quartic term is

$$\mathcal{O}_4(P) = [C, P] + \frac{1}{2} D^2 Q[P, P] - J P[Q, P] - \frac{1}{2} D^2 P[Q, Q]. \quad (\text{E.1})$$

Put

$$\Sigma_{-2} = \langle y^2, qy, ky, az, dz, ac, as, cd, ds, q^2, kq, k^2 \rangle$$

and

$$P_\sigma = -d^2 e_a + \sigma e_s \quad (\sigma \in \Sigma_{-2}).$$

Proposition E.1 (A positive-dimensional obstructed compression family). *For every $\sigma \in \Sigma_{-2}$, the cubic jet $C + [Q, P_\sigma]$ is supported only in the six output coordinates x, y, z, b, c, s , and therefore has coordinate-span rank at most six. Nevertheless,*

$$\Lambda_4(\mathcal{O}_4(P_\sigma)) = 1.$$

Hence no cubic source correction removes the quartic term anywhere in this 12-parameter formal family. Excluding the two monomials involving s gives a 10-parameter family of actual triangular polynomial automorphisms.

A larger exact calculation gives the following natural slice. Let

$$\begin{aligned} \mathcal{S} = \{P \in \mathfrak{X}_2^{(0)} : (C + [Q, P])_a = (C + [Q, P])_d = (C + [Q, P])_q = 0, \\ (C + [Q, P])_h = (C + [Q, P])_k = 0\}. \end{aligned}$$

Then $\mathcal{S}$ is a nonempty affine space of dimension 20, and

$$\Lambda_4(\mathcal{O}_4(P)) = 1 \quad (P \in \mathcal{S}).$$

The tangent space at $P_0 = -d^2e_a$ to the equivariant rank-at-most-six locus has dimension 22; the two extra directions are precisely the infinitesimal weight-preserving target changes that add the surviving y -row to the q - and k -rows. Thus $\mathcal{S}$ is the expected 20-dimensional transverse slice modulo linear target reparametrization.

This changes the interpretation of the obstruction: cubic compression is not isolated or difficult. It occurs in a family, while a secondary quartic class is constant and nonzero across the family.

AI AND COMPUTATIONAL DISCLOSURE

AI systems were used extensively in exploration, symbolic-program development, proof drafting, source organization, and manuscript editing. They are not authors. The public 11-variable source also discloses that its explicit construction and verification code were generated by ChatGPT [2]. The evidence for the computer-assisted theorems is the exact certificate suite described in section 7, not language model output. Human specialist review has not yet been completed.

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